Hai T.Phan

714-613-2959 • 27phanh@gmail.com • https://linkedin.com/in/HaiPhan • Garden Grove, CA
https://github.com/HaiPhan285

EDUCATION

California State University, Long Beach May 2028

- Bachelor of Science in Computer Engineering. GPA: 3.87

TECHNICAL SKILLS

- **Programming**: C, C++
- **Embedded Systems**: STM32, GPIO, Timers, PWM, ADC, Interrupts, PID Control
- **Hardware Design**: PCB Schematic Capture, PCB Layout, Motor Driver Circuits, Sensor Integration
- **Digital Design**: FPGA (Vivado)
- **Tools**: OrCAD PSpice, VS Code, Ubuntu Linux, Stm32, Git
- **Languages**: Vietnamese (Fluent)

PROJECTS

Micromouse Autonomous Maze-Solving Robot

- Develop embedded firmware in C++ for an STM32 microcontroller, leveraging strong debugging skills for real-time operation.
- Configure GPIO, timers, PWM, ADC, and interrupts for precise motor control and sensor interfacing.
- Implemented closed-loop PID control for DC motor speed regulation and accurate turning.
- Process analog wall-detection sensor data via ADC sampling for navigation decisions.
- Design a custom PCB schematic integrating an STM32 MCU, motor driver circuitry, sensor interfaces, and voltage regulation using schematic capture tools.
- Route PCB layout with careful consideration for signal path separation to minimize electrical noise.
- Perform board bring-up, firmware flashing, peripheral validation, and hardware–software debugging.

8-Bit Computer

- Designed digital schematics using Flux for a discrete 8-bit computer built from logic ICs.
- Implemented datapath components, including registers, ALU connections, and control signal routing.
- Developed instruction control logic to coordinate arithmetic operations and register transfers.
- Selected logic ICs based on functional requirements and signal compatibility.
- Built and wired the physical board from the schematic design.
- Tested and verified instruction execution through signal tracing and functional validation.
- Simulated critical logic sections using OrCAD PSpice before hardware implementation.

Compiler

- Implement a lexical scanner in C to tokenize arithmetic expressions, demonstrating proficiency in C/C++.
- Design token structures and enumerations for operators and integer literals.
- Built character-stream utilities with single-character lookahead.
- Implement multi-digit integer parsing with controlled character pushback.
- Add line-aware error reporting for invalid token detection.
- Structure code to support future recursive-descent parsing stages, showcasing debugging and code organization
- skills.

GrowBot – 3-Axis Autonomous Robot

- Developed motion control logic using C++ and microcontroller-based systems
- Integrated camera sensor input for object detection and automated pickup
- Coordinated synchronized control of three motion axes